



Savitribai Phule Pune University
Centre For Distance and Online Education

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
CENTRE FOR DISTANCE AND ONLINE EDUCATION

Program	Master of Computer Application	
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Topic Details

S.No	Topic	Pg.No
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OBJECTIVE

1. The objective of this content is to help students read and write functions in a direct and standard form.
2. In many math courses, students learn about functions early in their studies.
3. However, students often confuse the rule of a function, the name of the rule, and the output.
4. A direct notation system separates these parts clearly.
5. This separation helps the reader see the function name, the set of inputs, and the method to find the output in one line.
6. The main focus is on the symbols $f(x)$, $y = f(x)$, and $f: A \rightarrow B$.
7. These symbols are commonly used in undergraduate algebra, calculus, and data-based courses.
8. By the end of this content, a student should be able to complete five tasks.
9. First, the student should be able to define a function using words that explain the one output rule.
10. Second, the student should be able to write a function using direct notation with a name, a rule, and sets for inputs and outputs.
11. Third, the student should be able to evaluate a function at a given input, such as finding the value of $f(3)$.
12. Fourth, the student should be able to understand common types of functions, including linear, quadratic, and piecewise rules.
13. Fifth, the student should be able to connect function notation to real-life situations.
14. These situations may include a price rule or a speed rule.
15. The student should also be able to explain what the variables mean in these situations.

1.0 INTRODUCTION

1.1 Why Notation Matters

Mathematics is often described as a language because it uses symbols and patterns to share ideas clearly. Like any language, mathematics works well only when everyone follows the same rules for writing. Function notation is one of the most important rules in mathematics because functions appear in many topics. When notation is clear and direct, students can read textbooks, lab reports, or research papers and quickly understand relationships. In academic work, clear notation helps readers check steps and repeat results. For undergraduate students, correct notation also helps in solving problems because it keeps work neat and reduces mistakes.

1.1.1 Math As a Language

In everyday language, a word such as “tree” represents an idea or object. In mathematics, a symbol such as x represents a number or another mathematical object. Symbols alone do not have meaning unless definitions are given. Function notation is a short way to give such definitions. For example, writing $f(x)$ means taking an input x , applying a rule called f , and getting an output. This short form avoids long explanations and allows mathematical writing to remain clear, precise, and consistent.

1.2 The Idea of Input and Output

Many students first learn about functions through the idea of input and output. An input is a value that is chosen or measured. An output is the value that comes from applying a rule to that input. In science, inputs may include time, temperature, or mass, while outputs may include distance, pressure, or energy. In economics, the input might be the number of products sold, and the output might be the total cost. The function idea works well because it requires exactly one output for each input. If one input gave more than one output in the same situation, the rule would be unclear and unreliable.

2.0 MEANING AND DEFINITION

2.1 Meaning of a Function

A function is a rule, process, or mapping that assigns to each input exactly one output. The key words are “each” and “exactly one.” The rule can be described in words, a formula, a table, a graph, or computer code. The form is not the main point; the one output rule is the main point. When a student writes direct notation, the student is choosing a standard way to show the rule and to show how to use it. This is why function notation is both a content topic and a writing skill in mathematics.

Fig. 1

Functions in Mathematics: Meaning, Definition, and Related Terms

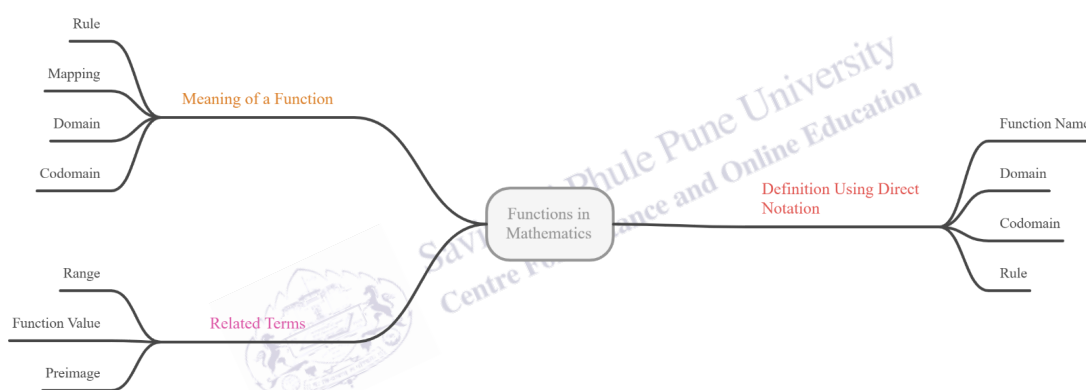


Diagram presents a mind map on Functions in Mathematics, showing meaning of a function, direct notation definition, and related terms. Branches include rule, mapping, domain, codomain, range, function value, preimage, and notation elements like function name and symbolic arrows relationships.

2.1.1 Functions as Mappings

In the mapping view, a function connects two sets. A set is a collection of objects, such as numbers, points, or strings in computer science. The first set is the domain, which contains allowed inputs. The second set is the codomain, which contains possible outputs. A function maps, or sends, every element of the domain to one element of the codomain. This view is common in higher level courses because it works for more than just real numbers. For example, a

function can map a student ID to a dorm room, or it can map a date to a daily average temperature.

2.2 Definition Using Direct Notation

A direct and complete way to denote a function is to state its name, its domain, its codomain, and its rule. A common academic format is: $f: A \rightarrow B$, defined by $f(x) = \text{expression for } x \text{ in } A$. In this line, the letter f names the function, the arrow shows direction from inputs to outputs, and the equation shows how to compute the output. The phrase “for x in A ” matters because it reminds the reader that not every symbol x is allowed. If the domain is not stated, confusion can happen, especially when a rule includes division by x or a square root.

2.3 Related Terms

Three related terms often appear with function notation. The first is the range, also called the image, which is the set of outputs the function actually produces from the domain. The range can be smaller than the codomain. The second is function value, which means a specific output like $f(3)$. The third is preimage, which means the inputs that lead to a given output. These terms support careful reading and writing. When students use them correctly, they can explain solutions in a way that matches academic standards and reduces ambiguity.



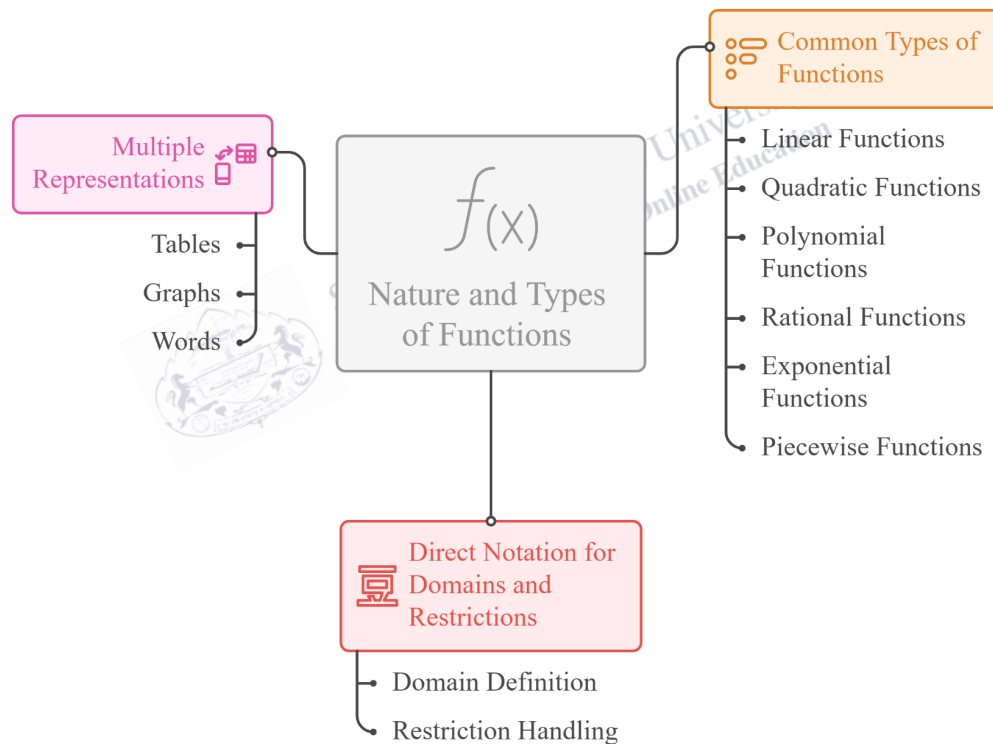
3.0 NATURE OR TYPE

3.1 Common Types of Functions

Functions come in types that share a common pattern. Types help students recognize what tools may work, such as how to graph, how to solve for inputs, or how the output changes. A type does not change the definition of a function; it describes the rule's structure. In undergraduate courses, common types include linear, quadratic, polynomial, rational, exponential, and piecewise functions. Learning to denote these types directly helps students communicate results and check reasonableness.

Fig. 2

Nature and Types of Functions



The diagram explains the nature and types of functions, showing common function types, multiple representations like tables and graphs, and direct notation for domains and restrictions. It highlights how functions are classified, represented, and formally defined in mathematics.

3.1.1 Linear and Quadratic Forms

A linear function has the form $f(x) = mx + b$, where m and b are constants. The graph is a straight line, and the output changes by a constant amount for each step in the input. A quadratic function has the form $f(x) = ax^2 + bx + c$, where a is not zero. The graph is a parabola, and the change in output is not constant. These two forms appear often, so the notation $f(x)$ makes it easy to compare them: the symbol x shows the input place, and the expression shows the rule.

3.2 Direct Notation for Domains and Restrictions

The nature of a function also includes where it is defined. Some rules work for all real numbers, but many do not. For example, the rule $g(x) = 1/x$ is not defined at $x = 0$, and the rule $h(x) = \sqrt{x}$ is not defined for negative real x . A direct way to denote such functions is to write the domain clearly, such as $g: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ or $h: [0, \infty) \rightarrow \mathbb{R}$. This style matches academic expectations because it states assumptions. It also helps readers avoid errors when they later solve equations like $g(x) = 5$ or evaluate $h(-1)$.

3.3 Multiple Representations

A function can be represented in several ways, and direct notation connects these ways. In a table, inputs appear in one column and outputs appear in another. In a graph, inputs appear on the horizontal axis and outputs on the vertical axis. In words, a rule may say, “Multiply the input by two, then add three.” A direct notation statement like $f(x) = 2x + 3$ links all of these. The student can compute table values, plot points, and describe the pattern using the same symbol f . In academic work, this consistency is important because it supports clear explanations in homework, exams, and reports.

4.0 EXAMPLES AND CASE STUDIES

4.1 Example: Writing a Function From Words

Suppose a rule says, “Take an input number, multiply it by 4, and then subtract 1.” A direct notation is $f(x) = 4x - 1$. If the domain is all real numbers, we can write $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 4x - 1$. This single line gives a complete description. To evaluate, compute $f(3) = 4 \cdot 3 - 1 = 11$. To check the function idea, note that the same input 3 always gives the same output 11, so the rule is stable.

4.2 Example: Function Table and Notation

Consider a small table with inputs 0, 1, 2, 3 and outputs 2, 5, 8, 11. The pattern is that the output increases by 3 each step. A linear rule fits: $f(x) = 3x + 2$. The notation explains the table because $f(0) = 2$, $f(1) = 5$, $f(2) = 8$, and $f(3) = 11$. If a table listed input 2 with two different outputs, such as 8 and 9, the table would not describe a function. In academic settings, students should state this clearly: the one output rule fails.

4.3 Example: Domain Matters

Let $g(x) = (x + 1)/(x - 2)$. The direct notation should include a domain restriction because the denominator is zero when $x = 2$. A clear statement is $g: \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R}, g(x) = (x + 1)/(x - 2)$. Now the reader knows that $g(2)$ is not allowed. If a student forgets this, the student might try to compute $g(2)$ and get an error. By writing the domain, the student shows careful academic practice, similar to stating conditions in a science method section.

4.4 Case Study: Mobile Data Plan Cost

A simple real world case is a mobile data plan. Assume a plan charges a fixed fee of 200 rupees each month plus 15 rupees for each gigabyte used. Let x be the number of gigabytes used in a month. Let $C(x)$ be the total cost. A direct function statement is $C: [0, \infty) \rightarrow \mathbb{R}, C(x) = 200 + 15x$. The domain starts at 0 because negative usage is not meaningful. This model helps explain how functions support decisions. If a student expects to use 10 GB, the model gives $C(10) = 200 + 150 = 350$ rupees. In academic writing, this example also shows good variable naming, because C reminds readers that the output is cost.

4.5 Case Study: Temperature Conversion

Another classic case is temperature conversion. Let $F(C)$ represent the Fahrenheit temperature that matches Celsius temperature C . The rule is $F(C) = (9/5)C + 32$. The notation uses C as the input symbol inside the parentheses to show that the input is Celsius. A direct way to present this in a report is: $F: \mathbb{R} \rightarrow \mathbb{R}, F(C) = (9/5)C + 32$. If $C = 0$, then $F(0) = 32$, which matches the freezing point of water in Fahrenheit. This function is linear, so the output changes by a constant factor

when the input changes. The notation is direct because it states the conversion rule and makes the role of each unit clear.

4.6 Case Study: A Piecewise Rule for Shipping

Some real situations need different rules in different ranges. For example, a store might charge 50 rupees shipping for orders below 500 rupees, and free shipping for orders of 500 rupees or more. Let x be the order amount. Define $S(x)$ as the shipping cost. A direct notation is: $S: [0, \infty) \rightarrow \mathbb{R}$, and $S(x) = 50$ if $0 \leq x < 500$, and $S(x) = 0$ if $x \geq 500$. This is a piecewise function because the rule depends on a condition. It is still a function because each order amount x has exactly one shipping cost. In undergraduate work, piecewise notation is common, so writing it clearly is part of academic communication.

1. A function is a rule that gives one output for each input.
2. Direct notation names the function and shows how to compute outputs.
3. Writing the domain prevents illegal substitutions and hidden errors.
4. The codomain is a target set, while the range is what actually appears.
5. The notation $f(x)$ is read as “ f of x ” and refers to the output when x is used as the input.
6. The notation $f: A \rightarrow B$ shows the mapping from a set of inputs to a set of outputs.
7. Tables and graphs are also valid ways to represent the same function.
8. Linear, quadratic, and piecewise functions are common types used in models.
9. Real-life cases such as cost and unit conversion show why function notation is important.
10. Academic mathematics writing values consistent symbols and clearly stated assumptions.



5.0 Keyword Touch Vocabulary Meaning

Keyword	Simple Meaning	Use in This Topic
Function	A rule from input to output	Main object being defined
Notation	A standard way to write	Makes math communication clear
Input	Value put into the rule	Often written as x
Output	Value produced by the rule	Often written as $f(x)$
Domain	Allowed inputs	Prevents invalid values
Codomain	Possible output set	Target set in $f: A \rightarrow B$
Range	Outputs that occur	Image of the domain
Evaluate	Compute a function value	Find $f(3)$ by substitution
Mapping	Pairing from one set to another	Arrow view of functions
Rule	Steps that make the output	Formula, words, or code
Variable	A symbol that can change	x or C inside parentheses
Constant	A fixed number	m , b , 32 , 200
Linear	Straight line type	$f(x)=mx+b$ models steady change
Quadratic	Has x squared	Parabola shaped graph
Rational	Ratio of polynomials	Needs domain restrictions
Exponential	Variable in exponent	Models growth and decay
Piecewise	Different cases	Uses conditions for the rule
Model	Math picture of reality	Uses functions to represent data
Preimage	Inputs leading to an output	Solving $f(x)=y$
Graph	Picture of input output	Shows how outputs change